

HIMEJI, Japan

WMO: 477690

Lat: 34.8394N

Lon: 134.6708E

Elev: 130

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.2	29.8	17.2	13.2	36.6	19.7	14.9	37.6	21.1	41.7	19.0	41.6	5.9	330	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.7	92.4	78.2	90.6	77.6	88.7	77.0	79.8	88.8	78.9	87.2	78.1	85.9	9.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.4	83.9	76.6	139.2	83.4	75.8	135.8	82.8	43.5	88.9	42.6	87.6	41.7	86.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.2	15.6	13.5	24.7	95.2	2.1	1.5	23.2	96.2	21.9	97.1	20.8	97.9	19.2	99.0		
			22.9	81.4	2.3	0.9	21.2	82.0	19.9	82.5	18.6	83.0	16.9	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	40.3	41.3	47.0	56.3	65.3	72.5	80.0	82.7	75.4	64.8	53.9	44.4
	DBStd	15.34	3.82	4.66	5.39	5.81	4.12	3.90	3.86	3.17	5.11	5.18	5.60	4.80
	HDD50	898	302	247	127	14	0	0	0	0	0	24	184	
	HDD65	3337	767	663	558	263	46	1	0	1	66	334	638	
	CDD50	4712	1	5	34	204	476	675	930	1013	763	460	140	11
	CDD65	1674	0	0	0	3	57	226	465	548	314	61	0	0
	CDH74	14925	0	0	0	16	264	1243	4474	6223	2456	248	1	0
	CDH80	5106	0	0	0	0	16	201	1570	2604	698	17	0	0
Wind	WSAvg	6.1	6.2	6.1	6.4	6.4	6.2	5.7	6.0	6.4	6.2	5.7	5.6	6.0
Precipitation	PrecAvg	49.6	2.0	2.4	3.8	3.9	5.7	5.9	7.0	4.0	6.0	4.2	2.8	2.0
	PrecMax	69.7	4.8	5.2	5.9	8.0	12.3	10.9	14.9	9.7	15.9	10.5	6.7	3.5
	PrecMin	30.6	0.6	0.7	1.7	0.9	2.2	2.2	0.8	0.4	1.7	0.8	0.4	0.4
	PrecStd	10.1	1.0	1.2	1.0	1.7	2.4	2.2	4.4	2.9	3.4	2.3	1.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	61.4	67.5	76.1	82.2	88.0	93.8	95.3	91.1	82.2	71.8	62.0
		MCWB	48.4	54.6	55.7	61.2	65.8	74.9	79.2	78.6	77.3	71.3	61.4	54.5
	2%	DB	52.9	56.7	63.4	72.4	79.2	84.3	91.5	93.3	88.5	79.0	68.9	58.7
		MCWB	45.8	49.7	54.0	59.3	64.6	72.3	78.6	78.2	75.8	68.9	60.0	51.8
	5%	DB	50.7	53.3	60.7	69.8	77.0	82.2	89.5	91.7	86.4	76.8	66.3	56.1
		MCWB	43.9	46.4	52.5	58.3	63.9	70.7	77.7	77.8	74.6	66.7	59.0	48.9
	10%	DB	48.4	50.6	57.9	67.2	74.8	80.3	87.5	90.1	84.2	74.5	63.9	53.5
		MCWB	42.2	44.1	50.4	56.9	62.9	69.8	77.0	77.4	73.4	65.0	56.8	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	56.8	59.9	66.2	71.0	77.2	81.2	81.0	79.7	74.8	65.3	58.1
		MCDB	54.4	59.7	62.8	69.7	75.1	83.8	90.4	90.8	86.8	79.2	67.8	59.9
	2%	WB	47.6	51.5	56.9	63.4	68.8	75.6	79.8	80.0	77.9	71.6	62.8	53.2
		MCDB	50.6	54.9	61.1	67.6	73.2	81.1	88.5	89.5	84.8	76.9	66.5	56.8
	5%	WB	44.9	48.0	54.1	61.3	67.2	74.2	78.8	79.1	76.8	69.0	60.4	50.3
		MCDB	49.7	51.8	58.6	66.7	72.6	79.1	86.8	87.9	83.2	74.5	64.5	54.4
	10%	WB	42.6	45.0	51.4	59.2	65.5	72.7	77.9	78.4	75.7	66.6	58.1	47.9
		MCDB	47.7	49.5	57.0	64.9	71.6	77.3	85.2	86.7	81.8	72.4	62.6	52.8
Mean Daily Temperature Range	5% DB	MDBR	15.3	15.9	17.4	17.9	16.8	13.5	12.1	13.7	14.3	15.8	16.3	15.5
		MCDBR	17.9	18.8	20.4	20.7	20.1	16.0	14.8	15.7	15.8	17.2	17.5	17.8
		MCWBR	11.6	12.7	13.0	11.1	9.7	7.0	5.7	5.5	6.6	8.8	10.7	11.5
	5% WB	MCDBR	16.3	15.9	16.8	15.2	14.3	11.5	13.4	14.0	12.7	13.3	14.5	15.6
		MCWBR	12.0	12.7	13.4	10.0	9.0	5.9	5.4	5.3	5.9	8.7	10.5	11.8
Clear-Sky Solar Irradiance	taub	0.405	0.446	0.509	0.538	0.543	0.565	0.568	0.539	0.490	0.447	0.420	0.399	
	taud	2.127	2.035	1.894	1.877	1.931	1.948	1.997	2.066	2.168	2.200	2.177	2.161	
	Ebn at Noon	243	247	241	242	243	237	235	239	245	245	237	237	
	Edh at Noon	39	47	59	63	60	59	56	51	44	40	36	35	
All-Sky Solar Radiation	RadAvg	697	897	1218	1516	1654	1510	1576	1630	1262	1028	760	629	
	RadStd	49	77	95	134	160	133	215	167	116	99	63	45	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page